

Project Title

Lightweight Person Re-Identification Model

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What is Person Re-Identification?

- Identifies the same person across different camera views without using facial recognition.
- Uses body appearance, clothing, and movement patterns.

Motivation

- Smart surveillance
- Public safety
- Airport and railway security
- Smart cities
- Retail analytics

Problem Statement

Existing Person Re-ID models achieve high accuracy but are computationally expensive, making them unsuitable for real-time deployment on edge devices. This project aims to develop a lightweight model that maintains high accuracy while reducing computational cost.

Base Paper

Title

DCR-ReID: Deep Component Reconstruction for Cloth-Changing Person Re-Identification

Problem Addressed

Traditional Person Re-ID models heavily rely on clothing appearance, making them ineffective when people change clothes. This paper focuses on Cloth-Changing Person Re-Identification (CC-ReID) by learning identity features that are independent of clothing.

Objective

Develop a lightweight Person Re-ID system using traditional feature extraction techniques instead of computationally expensive deep learning models.

Proposed Solution

The paper introduces DCR-ReID (Deep Component Reconstruction Re-ID), a framework that separates:

Clothes-Irrelevant Features

- Face
- Body Shape
- Body Structure
- Human Contour

Clothes-Relevant Features

- **Shirt**
- **Pants**
- **Color**
- **Texture**

This enables the model to recognize a person even after changing clothes.

Literature Review

Existing Method	Advantages	Limitations
LTCC (Dataset)	Realistic benchmark for cloth-changing person Re-ID	Provides only a dataset; no Re-ID model
FSAM (Fine-Grained Shape-Appearance Mutual Learning)	Learns body shape and appearance features jointly	Sensitive to pose variations and higher model complexity
CAL (Clothes-Changing Person Re-ID with RGB Only)	Learns clothing-invariant features using only RGB images	Incomplete separation of clothing and identity features
DCR-ReID (Base Paper)	Uses CRD and DAD modules for robust clothing-invariant feature learning; achieves state-of-the-art performance	Multi-branch architecture, higher computational cost, and requires more training resources

Research Gap

- Existing Re-ID models are highly dependent on clothing appearance.
- Performance decreases significantly when people change clothes or under challenging surveillance conditions.
- There is a need for a lightweight, clothing-invariant, and robust Person Re-ID model suitable for real-world applications.

Dataset Identification

Datasets Used

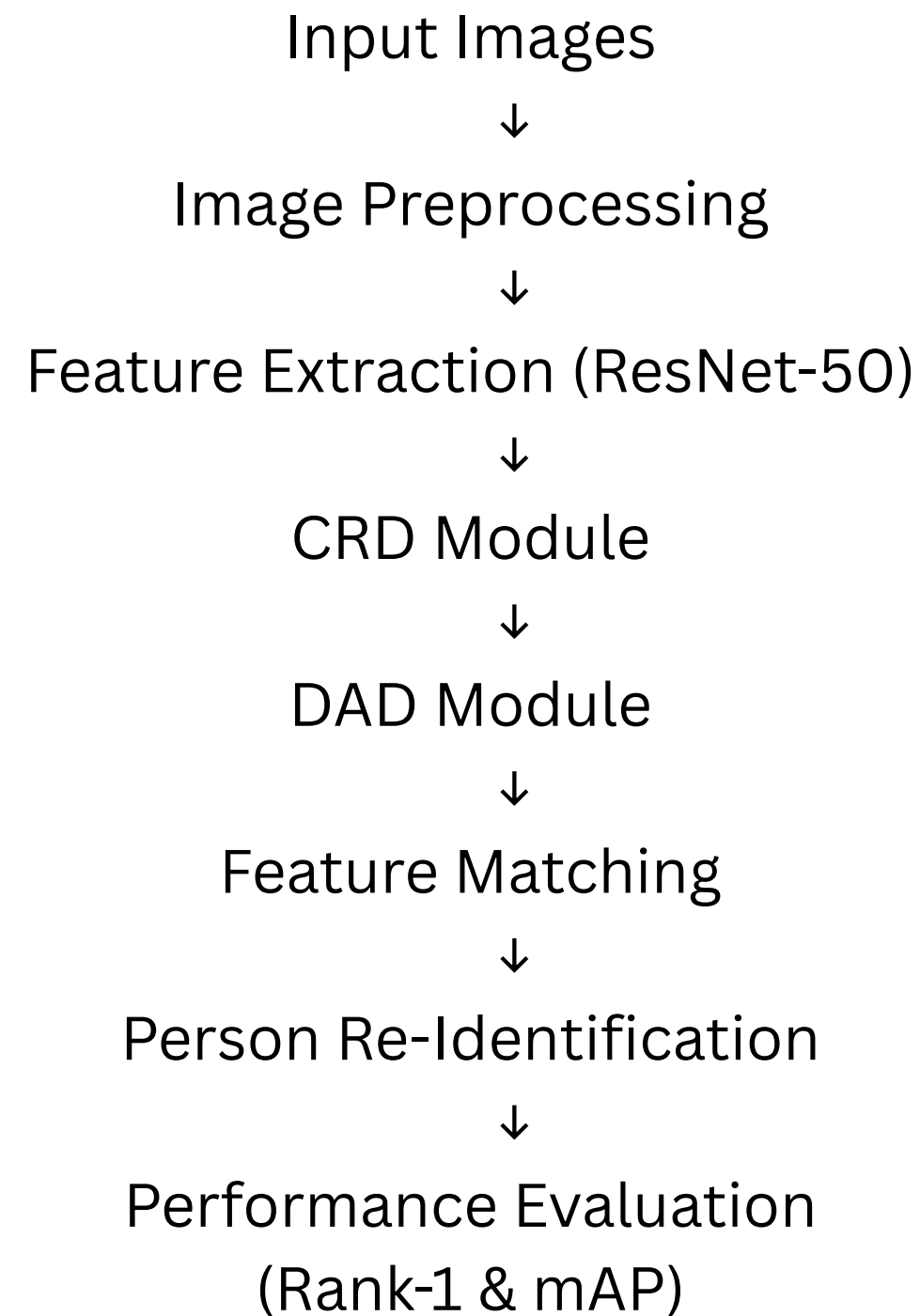
Kaggle Person Re-ID Dataset

- 684,287 images
- 1,500 identities
- Controlled environment

Challenges

- Occlusion
- Pose variation
- Illumination changes
- Background clutter
- Camera viewpoint changes

Workflow



Expected Outcome

- Develop a lightweight version of DCR-ReID with reduced computational complexity.
- Preserve high Rank-1 Accuracy and mAP while lowering model size and inference time.
- Improve robustness against clothing changes, occlusions, and viewpoint variations.
- Achieve a better balance between accuracy and computational efficiency for real-time surveillance applications.

Future Enhancement

- Develop a lightweight architecture to reduce computational complexity and improve inference speed.
- Enhance robustness against occlusions, pose variations, and challenging surveillance conditions.
- Enable real-time deployment on edge devices for practical surveillance applications.
- Extend the framework to support multi-modal Person Re-ID using RGB, thermal, and video data.
- Improve generalization across diverse datasets using domain adaptation and larger-scale training.